

Release Notes

Display Stream Compression (DSC) 1.2 Encoder

V2.5.1, February 16, 2022

Trilinear

Cores against chaos.

Introduction

This is the production release for the Trilinear Technologies Display Stream Compression Encoder Intellectual Property core. These release notes describe the features and enhancements for revision 2.5 of the core. These release notes also contain known information about supported technology families, implementations, and known issues and workarounds. The current update to the core provides several bug fixes and operational enhancements. Internal code updates have been made to guarantee proper post-synthesis operation with some toolsets.

Features

The Trilinear Technologies IP core has been optimized for both ASIC and FPGA implementations. The transmitter core features a technology abstraction layer which enables the controller to be easily ported to multiple implementation platforms. Each platform implementation may be optimized without requiring changes to the core logic.

Interfaces

The design includes three primary interfaces including the video input stream interface, the encoded video output interface and the AMBA APB-4 configuration interface. The encoder core does not perform time shift buffering of the input video data and has a limited capability to be stalled in a real-time system. Input data presented at the video input interface will begin encoding with minimal delay. Output data will be presented as soon as the encoded pixels are available. The APB-4 interface may be used to control the encode process and monitor the status of the core.

Delivery Types

The DSC Encoder Core may be delivered as a synthesizable soft core coded in the SystemVerilog hardware description language or as a technology specific netlist. Source code releases contain the entire design tree while netlist versions will contain a synthesized version of the core. Please refer to the DSC-E Integrator's Guide for additional information about specific delivery types.

Supported Technologies

The following families and specific devices are supported in this version. Please note that not all supported technologies will include support for all video input resolutions. Support for specific video modes is determined by the number of slice processors, the input data width and the frequency of the encoder clock.

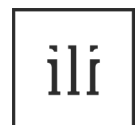
- Xilinx : UltraScale, Ultrascale+, Virtex7, Kintex7, Zynq
- Altera : Stratix V, Stratix 10
- TSMC : 65nm, 40nm LP, 28nm HP and HPM, 14nm, 12nm, 7nm
- SMIC : 40nm, 28nm, 18nm, 14nm
- UMC : 40nm, 28nm, 22nm, 14nm

NOTE: Xilinx Virtex-6 support will be discontinued as of February 1, 2022.

NOTE: TSMC 65nm and UMC 65nm support have been discontinued as of December 1, 2020.

NOTE: Altera Arria 10 support has been discontinued as of May 1, 2018.

NOTE: UMC 90nm support has been discontinued as of May 1, 2017.



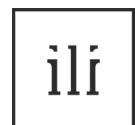
Supported Tool Flows

The current release of the DisplayPort Transmitter core has been tested using the following industry standard EDA tools.

- Synopsys Design Compiler Ultra 2020 and later
- Xilinx Vivado 2018.1 and later
- Intel Quartus Prime 18 and later
- Synopsys VCS 2020 and later
- Mentor ModelSim/Questa 2020 and later
- Cadence Xcelium 20.03 and later

Supported Test Environment

A Bus Functional Model (BFM) test environment is supported for the user level verification of the core using a SystemVerilog design flow. Only the SystemVerilog constructs supported as part of the “SystemVerilog for Design” subset are used in the implementation of the test environment.



Issues Fixed in This Release

Revision 2.5.1

DR#35 – Corrected an issue in the calculation of the flatness decision for ICH decision rejection. This case was not being covered in the compliance tests. A test was added to the regression list to cover this case. The flatness logic was also refactored for clarity and performance.

DR#34 – Fixed an issue where data could get stranded in the internal slice buffers for odd resolution input modes.

DR#33 – Primary Qp values were being constrained to the Min and Max Qp values when in DSC 1.1 mode. The Qp logic has been updated to account for the version number.

Revision 2.5.0

DR#32 – Corrected a possible issue that would allow the final, unused pixel in a group to be coded through VLC resulting in an invalid output stream.

DR#31 – Fixed an issue with the generation of the *next_flatness_flag* in some low compression rate applications. Added logic to properly detect and generate these flags under all conditions.

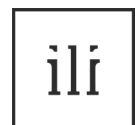
DR#30 – Found and fixed an issue in the output stream buffers that could cause a significant delay before generating a valid stream.

DR#29 – Fixed an issue in the slice output formatting block that could cause incorrect prefix sequences to be added to the bit stream in 12 and 16 bpc modes.

DR#28 – Corrected an issue in the output stream formatting block that could cause slice Co and Cg data to be swapped for first mux word at high compression rates.

DR#27 – Added logic to test for the ICH inhibit condition when configured for 16 bits per component on the input.

DR#26 – Fixed an issue in the VLC generator that could cause an incorrect bit sequence to be generated in the Y channel under some conditions.



New Features in This Release

Revision 2.5.1

ER#29 – Made minor corrections to the source code to comply with a stricter set of linting rules.

ER#28 – Reduced the size of the SPC slice buffer by 50%. The updated design combines the input buffer and the slice buffer. The input buffer has been completely removed. The enhancement has also improved performance for lines containing multiple slices.

ER#27 – Updated the source code to reflect the terminology changes in the DSC 1.2b specification.

ER#26 – Separated the group flatness calculation logic from the Qp update in the refactored flatness logic. Added the `dsce_rate_adjust.sv` module.

Revision 2.5.0

ER#25 – The encoding of the `AXI_OUTPUT_MODE` register has changed. Support for the 128-bit AXI bus output has been removed. Please see the User's Guide for additional details.

ER#24 – Refactored the output formatting code into a dedicated block named `dsce_stream_builder.sv`. This block now encapsulates the output stream assembly logic into one file. Proper muxword ordering is now strictly enforced.

ER#23 – Removed the ability to output the picture parameter set along with the compressed byte stream. This feature was only used for comparison to DSC verification files and unnecessarily increased the gate count of the core.

ER#22 – The programmable quantization level tables have been deprecated in favor of static lookup tables. This modification has improved timing and reduced the gate count of the design.

ER#21 – The Indexed Color History (ICH) function has been completely restructured to reduce gate count and improve performance.

